



Lecture 7 and 8

Lecture 7: Equilibrium Electrochemistry

**Lecture 8: Impact of Equilibria on
Engineering**



Equilibrium Electrochemistry

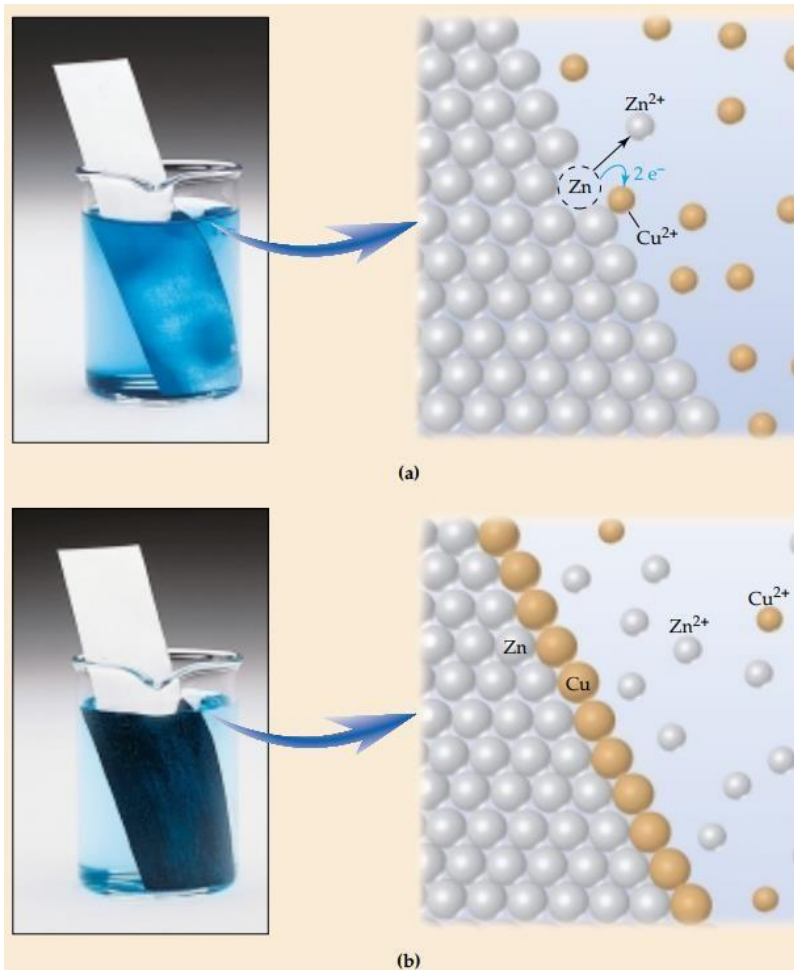


Electrochemistry: A branch of chemistry concerned with the interconversion of chemical and electrical energy

Galvanic Cell

Consider;

- (a) A strip of zinc **metal is immersed** in an aqueous CuSO_4 solution. The redox reaction takes place at the metal–solution interface and directly transfers two electrons from Zn atoms to Cu^{2+} ions.
- (b) **As time passes**, a dark colored deposit of copper metal appears on the zinc, and the blue color due to Cu^{2+} fades from the solution.

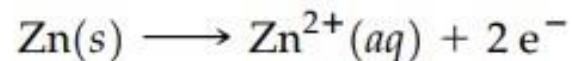




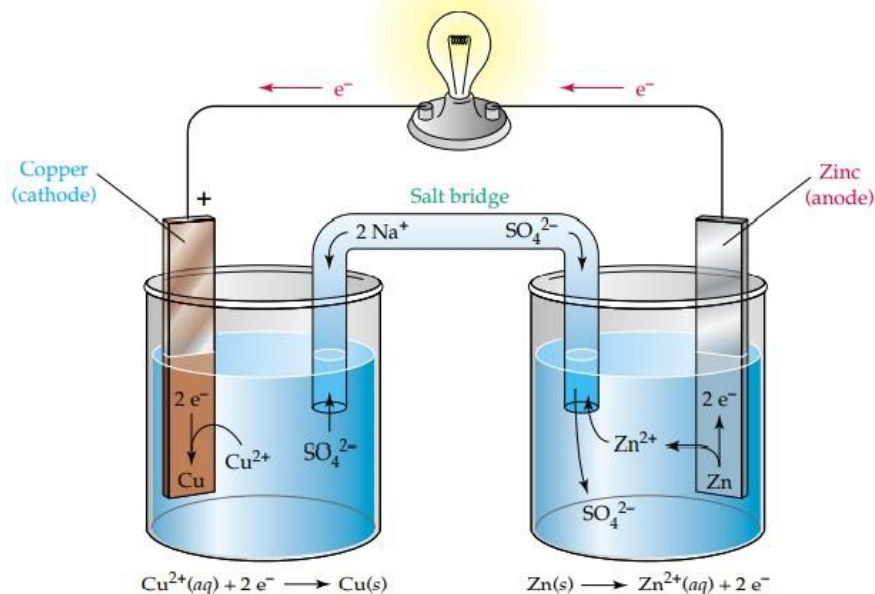
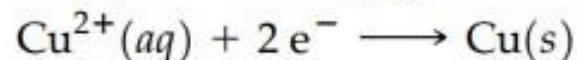
This is a redox reaction in which Zn (reducing agent) is oxidized to Zn^{2+} and Cu^{2+} (oxidizing agent) is reduced to Cu.

Recall: Oxidation is a **loss** of electrons and reduction is a **gain** of electrons. In form of half reactions;

Oxidation half-reaction:



Reduction half-reaction:



(a) A galvanic cell that uses oxidation of Zn to Zn^{2+} and reduction of Cu^{2+} to Cu metal. Note that the **negative particles (electrons in the wire and anions in solution) travel around the circuit in the same direction**. The resulting electric current can be used to light a lightbulb.



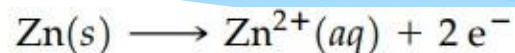
(b) An operating Daniell cell. The salt bridge in part (a) is replaced by a porous glass disk that allows ions to flow between the **anode and cathode** compartments but prevents bulk mixing, which would bring ions into direct contact with zinc and **short-circuit** the cell. The lightbulb in part (a) is replaced with a digital voltmeter



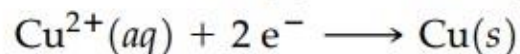
The electrode at which oxidation takes place is called **the anode** (the zinc strip in our example), and the electrode at which reduction takes place is called the **cathode** (the copper strip). The anode and cathode half-reactions must add to give the overall cell reaction:



Anode (oxidation) half-reaction:



Cathode (reduction) half-reaction:



Overall cell reaction:



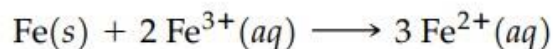
In the commercial world;

Anode: {
Is where oxidation occurs
Is where electrons are produced
Is what anions migrate toward
Has a negative sign

Cathode: {
Is where reduction occurs
Is where electrons are consumed
Is what cations migrate toward
Has a positive sign



Design a galvanic cell that uses the redox reaction



Self-test

Identify the anode and cathode half-reactions, and sketch the experimental setup. Label the anode and cathode, indicate the direction of electron and ion flow, and identify the sign of each electrode.

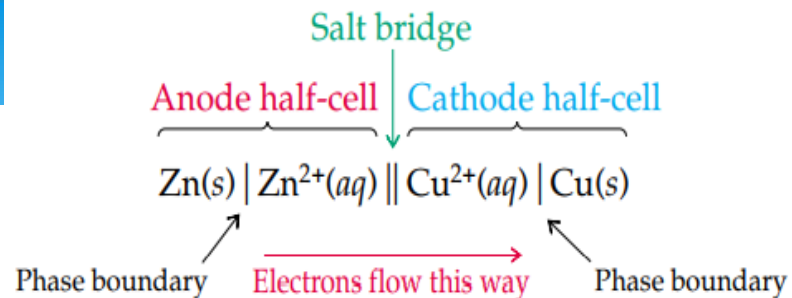
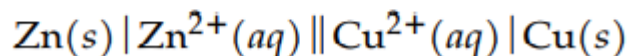
Shorthand Notation for Galvanic Cells



For the previous example,

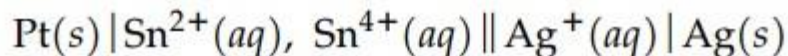


the SHN is



Example

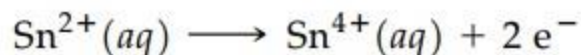
Given the following shorthand notation



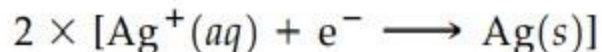
write a balanced equation for the cell reaction, and give a brief description of the cell.

Strategy

Because the anode always appears at the left in the shorthand notation, the anode (oxidation) half-reaction is



The cathode (reduction) half-reaction is



We multiply the cathode half-reaction by a factor of 2 so that the electrons will cancel when we sum the two half-reactions to give the cell reaction:





Free-Energy Changes for Cell Reactions



The **driving force that pushes the negatively charged electrons** away from the anode and pulls them toward the cathode is an electrical potential called the electromotive force (emf), also known as the cell potential (E) or the cell voltage. The SI unit is the volt (V), and the potential of a galvanic cell is defined as a positive quantity.

Two quantitative measures of the driving force of a chemical reaction: the **cell potential E (an electrochemical quantity)** and the **free-energy change (a thermochemical quantity)**. The values of ΔG and E are directly proportional and are related by the equation;

$$\Delta G = -nFE$$

where n is the number of moles of electrons transferred in the reaction and F is the faraday (or Faraday constant). E° and ΔG° refer to cell potential and **Gibbs free energy, respectively**, at standard conditions, related by:

$$\Delta G^\circ = -nFE^\circ$$

Calculate the standard free-energy change at 25°C for the following reaction. The standard cell potential is 1.10 V at 25°C.



To calculate ΔG° , we use the equation $\Delta G^\circ = -nFE^\circ$, where E° is given and n can be inferred from the balanced chemical equation.

SOLUTION

Two moles of electrons are transferred from Zn to Cu^{2+} in this reaction, and the standard free-energy change is therefore

$$\begin{aligned}\Delta G^\circ &= -nFE^\circ = -(2 \text{ mol e}^-) \left(\frac{96,500 \text{ C}}{\text{mol e}^-} \right) (1.10 \text{ V}) \left(\frac{1 \text{ J}}{1 \text{ C} \cdot \text{V}} \right) \\ &= -212,000 \text{ J} = -212 \text{ kJ}\end{aligned}$$

Self-test quiz:

The standard cell potential at 25°C is 0.92 V for the reaction below. What is the standard free-energy change for this reaction at 25°C?



Standard Reduction Potentials

The standard potential of any galvanic cell is the sum of the standard half-cell potentials for oxidation at the anode and reduction at the cathode:

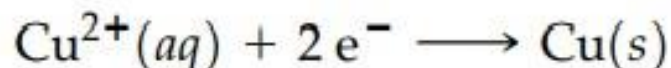
$$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{ox}} + E^{\circ}_{\text{red}}$$

Consider, for example, a cell in which H_2 gas is oxidized to H^+ ions at the anode and Cu^{2+} ions are reduced to copper metal at the cathode

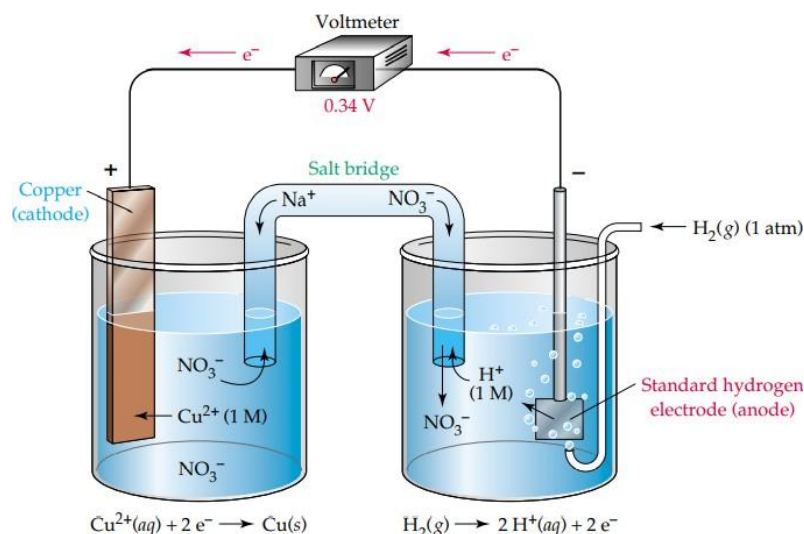
Anode (oxidation):



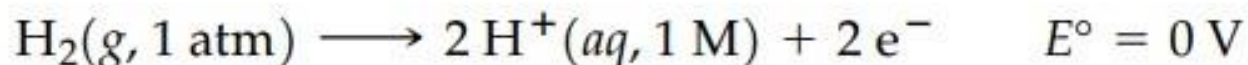
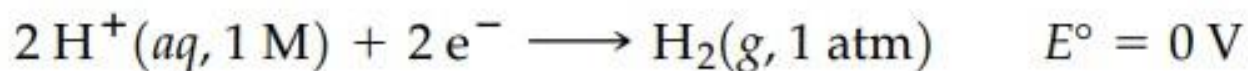
Cathode (reduction):



Overall cell reaction:



A galvanic cell consisting of a Cu^{2+} (1M)/Cu half-cell and a standard hydrogen electrode (S.H.E.). The S.H.E. is a piece of platinum foil that is in contact with bubbles of H_2 at 1 atm pressure and with H^+ at 1 M concentration. Electrons flow from the S.H.E. (anode) to the copper cathode. The measured standard cell potential at 25°C is 0.34 V.



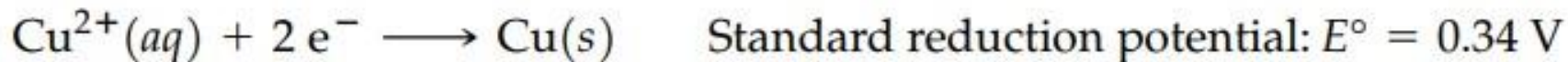
With this choice of standard reference electrode, the entire potential of the cell



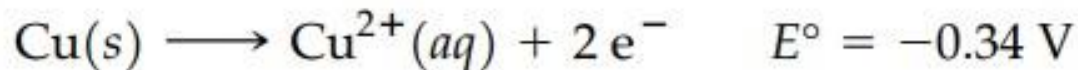
can be attributed to the Cu^{2+}/Cu half-cell:

$$\begin{array}{ccccc} E^\circ_{\text{cell}} = E^\circ_{\text{H}_2 \rightarrow \text{H}^+} + E^\circ_{\text{Cu}^{2+} \rightarrow \text{Cu}} = 0.34 \text{ V} \\ \uparrow \quad \quad \quad \uparrow \quad \quad \quad \uparrow \\ 0.34 \text{ V} \quad \quad 0 \text{ V} \quad \quad \quad 0.34 \text{ V} \end{array}$$

Because the Cu^{2+}/Cu half-reaction is a reduction, the corresponding half-cell potential, $E^\circ = 0.34 \text{ V}$, is called a **standard reduction potential**:



In a cell in which this half-reaction occurs in the opposite direction, the corresponding half-cell potential has the same magnitude but opposite sign:



Whenever the direction of a half-reaction is reversed, the sign of E° must also be reversed. Thus, the standard potential for an oxidation half-reaction is the negative of the standard reduction potential.

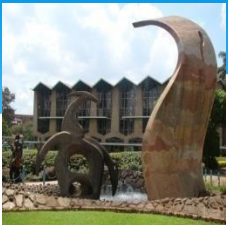
Standard Reduction Potentials at 25°C



Reduction Half-Reaction			E° (V)	
Stronger oxidizing agent	$F_2(g) + 2 e^-$	$\longrightarrow 2 F(aq)$	2.87	Weaker reducing agent
	$H_2O_2(aq) + 2 H^+(aq) + 2 e^-$	$\longrightarrow 2 H_2O(l)$	1.78	
	$MnO_4^-(aq) + 8 H^+(aq) + 5 e^-$	$\longrightarrow Mn^{2+}(aq) + 4 H_2O(l)$	1.51	
	$Cl_2(g) + 2 e^-$	$\longrightarrow 2 Cl^-(aq)$	1.36	
	$Cr_2O_7^{2-}(aq) + 14 H^+(aq) + 6 e^-$	$\longrightarrow 2 Cr^{3+}(aq) + 7 H_2O(l)$	1.33	
	$O_2(g) + 4 H^+(aq) + 4 e^-$	$\longrightarrow 2 H_2O(l)$	1.23	
	$Br_2(l) + 2 e^-$	$\longrightarrow 2 Br^-(aq)$	1.09	
	$Ag^+(aq) + e^-$	$\longrightarrow Ag(s)$	0.80	
	$Fe^{3+}(aq) + e^-$	$\longrightarrow Fe^{2+}(aq)$	0.77	
	$O_2(g) + 2 H^+(aq) + 2 e^-$	$\longrightarrow H_2O_2(aq)$	0.70	
	$I_2(s) + 2 e^-$	$\longrightarrow 2 I^-(aq)$	0.54	
	$O_2(g) + 2 H_2O(l) + 4 e^-$	$\longrightarrow 4 OH^-(aq)$	0.40	
	$Cu^{2+}(aq) + 2 e^-$	$\longrightarrow Cu(s)$	0.34	
	$Sn^{4+}(aq) + 2 e^-$	$\longrightarrow Sn^{2+}(aq)$	0.15	

The standard potential for the following galvanic cell is 0.92 V: $Al(s) | Al^{3+}(aq) || Cr^{3+}(aq) | Cr(s)$

Look up the standard reduction potential for the Al^{3+}/Al half-cell in Table
calculate the standard reduction potential for the Cr^{3+}/Cr half-cell.



The Nernst Equation



Cell potentials, like free-energy changes, **depend on temperature and on the composition of the reaction mixture**—that is, on the concentrations of solutes and the partial pressures of gases.

This dependence can be derived from the equation

$$\Delta G = \Delta G^\circ + RT \ln Q$$

ΔG is the free-energy change for a reaction under **nonstandard-state conditions**, ΔG° is the free-energy change under **standard-state conditions**, and Q is the reaction quotient. Since $\Delta G = -nFE$ and $\Delta G^\circ = -nFE^\circ$ we can rewrite the equation for in the form

$$-nFE = -nFE^\circ + RT \ln Q$$

Dividing by $-nF$, we obtain the **Nernst equation**, named after Walther Nernst (1864–1941), the German chemist who first derived it:

 **NERNST EQUATION** $E = E^\circ - \frac{RT}{nF} \ln Q$ or $E = E^\circ - \frac{2.303RT}{nF} \log Q$

At 25°C, $2.303RT/F = 0.0592$ V, and therefore;

$$E = E^\circ - \frac{0.0592 \text{ V}}{n} \log Q \quad \text{in volts, at 25°C}$$

Consider a galvanic cell that uses the reaction



Calculate the cell potential at 25°C when $[\text{H}^+] = 1.0 \text{ M}$, $[\text{Zn}^{2+}] = 0.0010 \text{ M}$, and $P_{\text{H}_2} = 0.10 \text{ atm}$.

SOLUTION

The standard cell potential is

$$E^\circ = E^\circ_{\text{Zn} \rightarrow \text{Zn}^{2+}} + E^\circ_{\text{H}^+ \rightarrow \text{H}_2} = -(-0.76 \text{ V}) + 0 \text{ V} = 0.76 \text{ V}$$

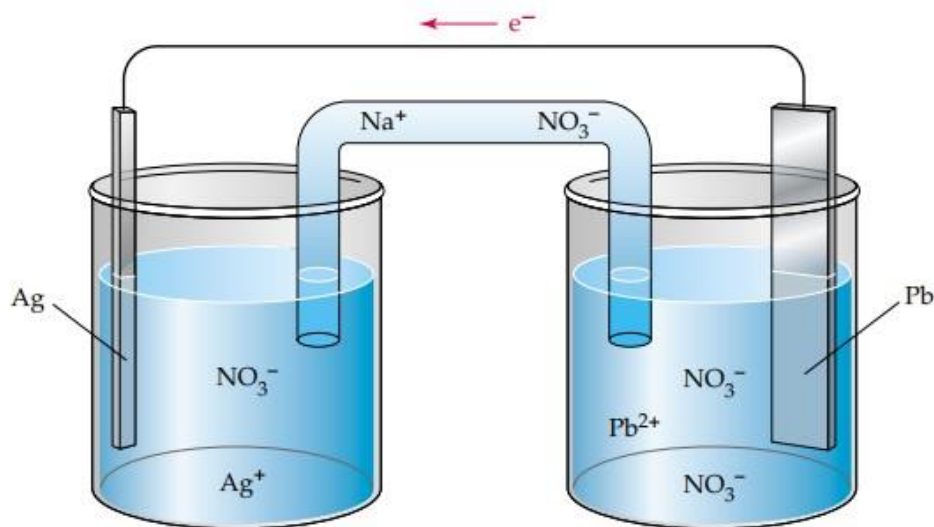
The cell potential at 25°C under nonstandard-state conditions is given by the Nernst equation:

$$\begin{aligned} E &= E^\circ - \frac{0.0592 \text{ V}}{n} \log Q \\ &= E^\circ - \left(\frac{0.0592 \text{ V}}{n} \right) \left(\log \frac{[\text{Zn}^{2+}](P_{\text{H}_2})}{[\text{H}^+]^2} \right) \end{aligned}$$

where the reaction quotient contains both molar concentrations of solutes and the partial pressure of a gas (in atm). As usual, zinc has been omitted from the reaction quotient because it's a pure solid. For this reaction, 2 mol of electrons are transferred, so $n = 2$. Substituting into the Nernst equation the appropriate values of E° , n , $[\text{H}^+]$, $[\text{Zn}^{2+}]$, and P_{H_2} gives

$$\begin{aligned} E &= (0.76 \text{ V}) - \left(\frac{0.0592 \text{ V}}{2} \right) \left(\log \frac{(0.0010)(0.10)}{(1.0)^2} \right) = (0.76 \text{ V}) - \left(\frac{0.0592 \text{ V}}{2} \right) (-4.0) \\ &= 0.76 \text{ V} + 0.12 \text{ V} \\ &= 0.88 \text{ V} \quad \text{at } 25^\circ\text{C} \end{aligned}$$

Consider the following galvanic cell:



STRATEGY

The direction of electron flow in the picture tells us that lead is the anode and silver is the cathode. Therefore, the cell reaction is $\text{Pb}(s) + 2 \text{Ag}^+(aq) \rightarrow \text{Pb}^{2+}(aq) + 2 \text{Ag}(s)$. The cell potential at 25°C is given by the Nernst equation, where $n = 2$ and $Q = [\text{Pb}^{2+}]/[\text{Ag}^+]^2$:

$$\begin{aligned} E &= E^\circ - \frac{0.0592 \text{ V}}{n} \log Q \\ &= E^\circ - \left(\frac{0.0592 \text{ V}}{2} \right) \left(\log \frac{[\text{Pb}^{2+}]}{[\text{Ag}^+]^2} \right) \end{aligned}$$

The change in E on changing the ion concentrations will be determined by the change in the log term in the Nernst equation.

SOLUTION

(a) Pb^{2+} is in the anode compartment, and Ag^+ is in the cathode compartment. Suppose that the original concentrations of Pb^{2+} and Ag^+ are 1 M, so that $E = E^\circ$. Increasing $[\text{Pb}^{2+}]$ to 10 M gives

$$E = E^\circ - \left(\frac{0.0592 \text{ V}}{2} \right) \left(\log \frac{(10)}{(1)^2} \right)$$

Because $\log 10 = 1.0$, $E = E^\circ - 0.03 \text{ V}$. Thus, increasing the Pb^{2+} concentration by a factor of 10 decreases the cell voltage by 0.03 V.

(b) Increasing $[\text{Ag}^+]$ to 10 M gives

$$E = E^\circ - \left(\frac{0.0592 \text{ V}}{2} \right) \left(\log \frac{(1)}{(10)^2} \right)$$

Because $\log (10)^{-2} = -2.0$, $E = E^\circ + 0.06 \text{ V}$. Thus, increasing the Ag^+ concentration by a factor of 10 increases the cell voltage by 0.06 V.



| Electrochemical Determination of pH

The electrochemical determination of pH using a pH meter is a particularly important application of the Nernst equation. Consider, for example, a cell with a hydrogen electrode as the anode and a second reference electrode as the cathode:



The hydrogen electrode consists of a platinum wire that is in contact with H_2 at 1 atm and dips into a solution of unknown pH. The potential of this cell is

$$E_{\text{cell}} = E_{\text{H}_2 \rightarrow \text{H}^+} + E_{\text{ref}}$$

We can calculate the potential for the hydrogen electrode half-reaction



by applying the Nernst equation to this half-reaction:

$$E_{\text{H}_2 \rightarrow \text{H}^+} = (E^\circ_{\text{H}_2 \rightarrow \text{H}^+}) - \left(\frac{0.0592 \text{ V}}{n} \right) \left(\log \frac{[\text{H}^+]^2}{P_{\text{H}_2}} \right)$$

Since $E^\circ = 0 \text{ V}$ for the standard hydrogen electrode, $n = 2$, and $P_{\text{H}_2} = 1 \text{ atm}$, we can rewrite this equation as

$$E_{\text{H}_2 \rightarrow \text{H}^+} = - \left(\frac{0.0592 \text{ V}}{2} \right) (\log [\text{H}^+]^2)$$

Further, because $\log [H^+]^2 = 2 \log [H^+]$ and $-\log [H^+] = \text{pH}$, the half-cell potential for the hydrogen electrode is directly proportional to the pH:

$$\begin{aligned} E_{H_2 \rightarrow H^+} &= -\left(\frac{0.0592 \text{ V}}{2}\right)(2)(\log [H^+]) = (0.0592 \text{ V})(-\log [H^+]) \\ &= (0.0592 \text{ V})(\text{pH}) \end{aligned}$$

The overall cell potential is

$$E_{\text{cell}} = (0.0592 \text{ V})(\text{pH}) + E_{\text{ref}}$$

and the pH is therefore a linear function of the cell potential:

$$\text{pH} = \frac{E_{\text{cell}} - E_{\text{ref}}}{0.0592 \text{ V}}$$

A higher cell potential indicates a higher pH, meaning that we can measure the pH of a solution simply by measuring E_{cell} .



Standard Cell Potentials and Equilibrium Constants



We have seen that the standard free-energy change for a reaction is related to the standard cell potential by the equation:

$$\Delta G^\circ = -nFE^\circ$$

In addition, the standard free-energy change is also related to the equilibrium constant for the reaction:

$$\Delta G^\circ = -RT \ln K$$

Combining the two: $-nFE^\circ = -RT \ln K$

Thus
$$E^\circ = \frac{RT}{nF} \ln K = \frac{2.303RT}{nF} \log K \quad E^\circ = \frac{0.0592 \text{ V}}{n} \log K \quad \text{in volts, at } 25^\circ\text{C}$$

As an example, let's calculate the value of K for the reaction in the Daniell cell:



Solving the equation for $\log K$ and substituting in the appropriate values of E° and n , we obtain

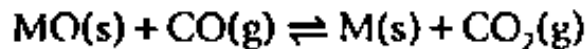
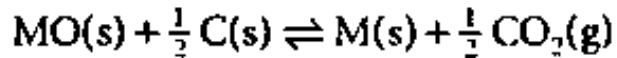
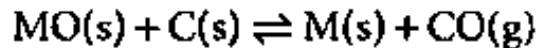
$$E^\circ = \frac{0.0592 \text{ V}}{n} \log K$$
$$\log K = \frac{nE^\circ}{0.0592 \text{ V}} = \frac{(2)(1.10 \text{ V})}{0.0592 \text{ V}} = 37.2$$

Lecture 8

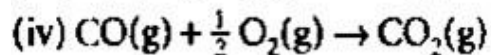
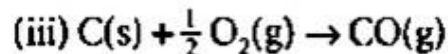
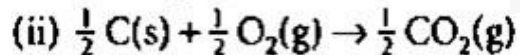
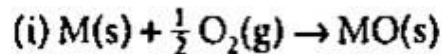
Impact of Equilibria on Engineering

The extraction of metals from their oxides

Metals can be obtained from their oxides by reduction with carbon or carbon monoxide if any of the equilibria lie to the right (that is, have $K > 1$).

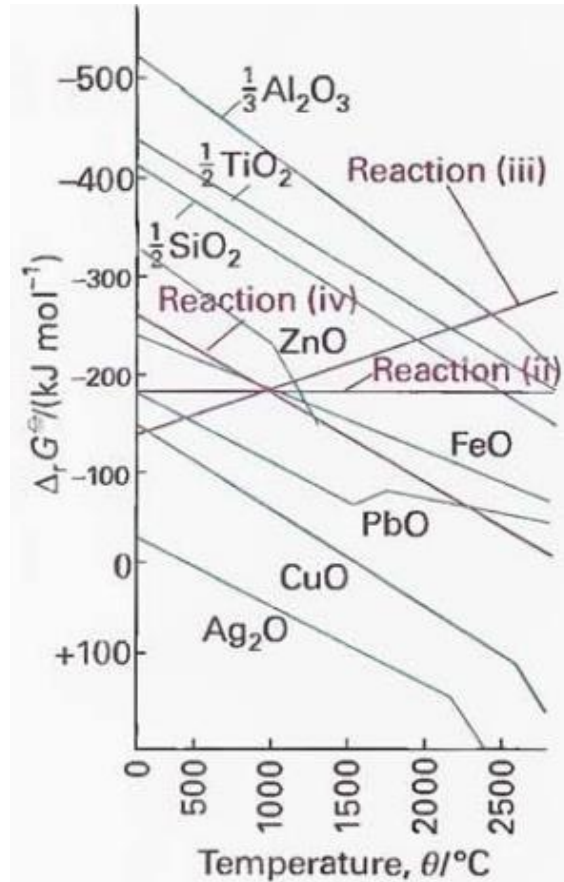


As we shall see, these equilibria can be discussed in terms of the thermodynamic functions for the reactions



Successful reduction of the oxide depends on the outcome of the competition of the carbon for the oxygen bound to the metal.

The temperature dependences of the standard Gibbs energies of reactions (i)–(iv) depend on the reaction entropy through $d\Delta_r G^\circ/dT = -\Delta_r S^\circ$. Because in reaction (iii) there is a net increase in the amount of gas, the standard reaction entropy is large and positive; therefore, its $\Delta_r G^\circ$ decreases sharply with increasing temperature. In reaction (iv), there is a similar net decrease in the amount of gas, so $\Delta_r G^\circ$ increases sharply with increasing temperature. In reaction (ii), the amount of gas is constant, so the entropy change is small and $\Delta_r G^\circ$ changes only slightly with temperature.



An Ellingham diagram for the discussion of the reduction of metal ores.



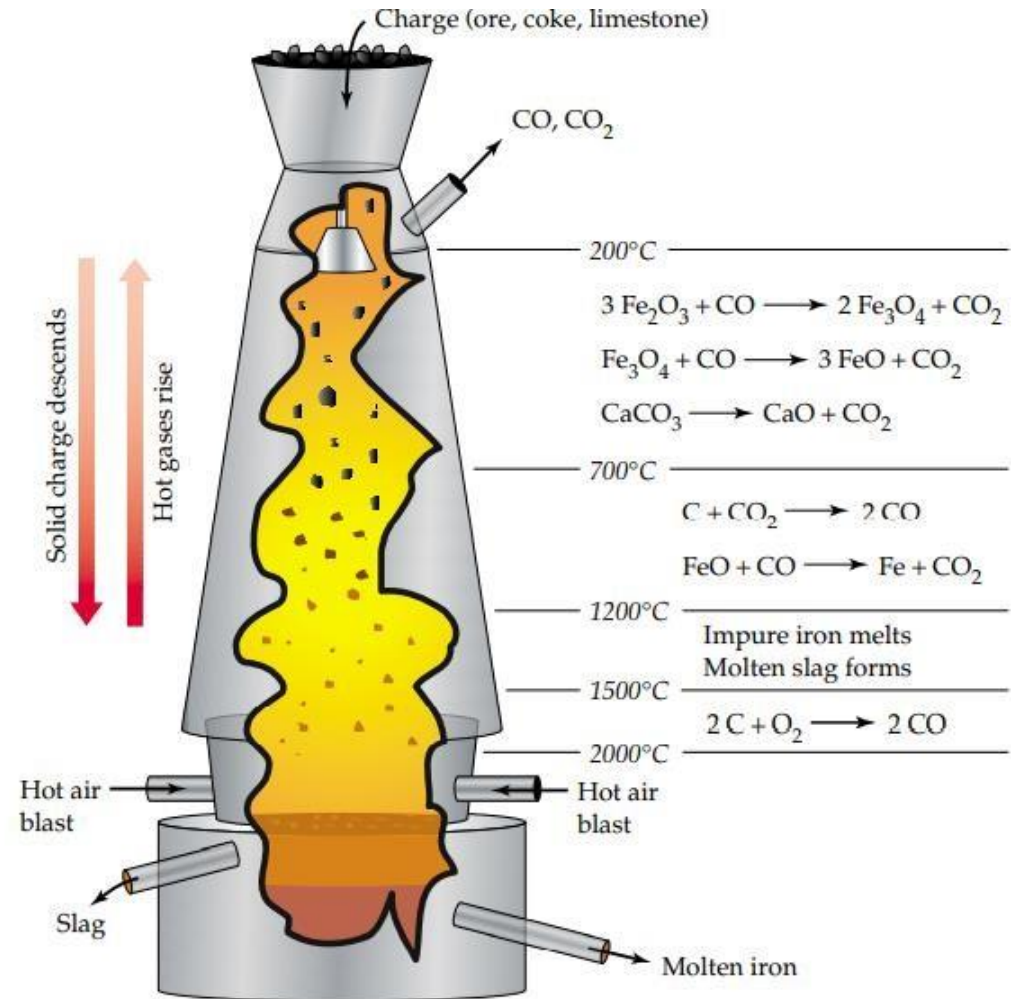
Example: Iron and Steel

The metallurgy of iron is technological importance because iron is the major constituent of steel, the most widely used.



Worldwide production of steel amounts to some 800 million tons per year. Iron is produced by CO reduction of iron ore Fe_2O_3 , usually hematite in a huge reactor called a **blast furnace**

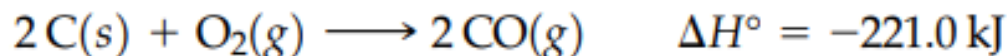
A charge of iron ore, coke, and limestone (CaCO_3) is introduced at the top of the furnace





Notes

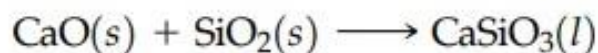
A blast of hot air is sent in at the bottom, burning coke and yielding carbon monoxide at a temperature of about 2000°C:



As the charge descends and the hot CO rises, a complex series of high-temperature reactions occur in the various regions of the furnace, as shown in Figure. The key overall reaction is the reduction of Fe_2O_3 to iron metal (MP 1538°C), which is obtained as an impure liquid at the bottom of the furnace:



The purpose of the limestone is to remove the gangue from the iron ore. At the high temperatures of the furnace, the limestone decomposes yielding lime (CaO), a basic oxide that reacts with SiO_2 and other acidic oxides present in the gangue. The product, called **slag**, is a molten material consisting mainly of calcium silicate:



Lime

Sand

Slag

The slag, which is less dense than the molten iron, floats on the surface of the iron, thus allowing the iron and the slag to be removed from the bottom of the furnace through separate taps.

The iron obtained from a blast furnace is a brittle material called *cast iron*, or *pig iron*.



Limestone and Cement Chemistry

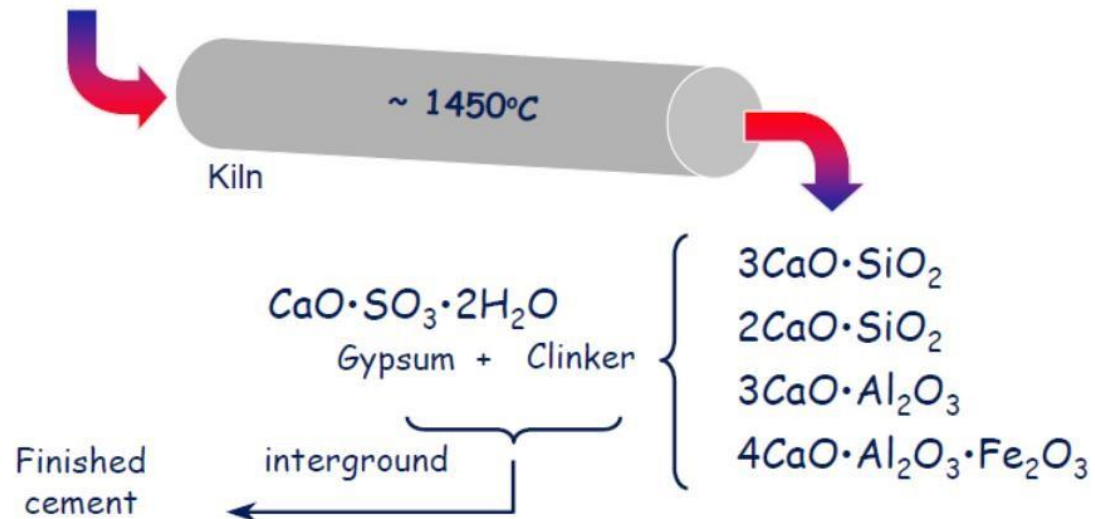


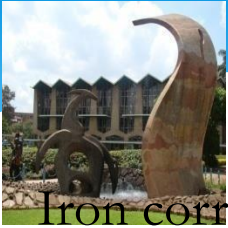
Limestone and rocks such as clay or shale contain lime (CaCO_3), silica (SiO_2), alumina (Al_2O_3) and ferrous oxide (Fe_2O_3) - the raw materials of cement manufacture.

The raw materials are then dried, heated and fed into a rotating kiln. Here the raw materials react at very high temperatures to form $3\text{CaO} \cdot \text{SiO}_2$ (tricalcium silicate), $2\text{CaO} \cdot \text{SiO}_2$ (dicalcium silicate), $3\text{CaO} \cdot \text{Al}_2\text{O}_3$ (tricalcium aluminate) and $4\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{Fe}_2\text{O}_3$ (tetracalcium alumino-ferrate).

Mineral	Chemical formula	Oxide composition	Abbreviation
Tricalcium silicate (alite)	Ca_3SiO_5	$3\text{CaO} \cdot \text{SiO}_2$	C3S
Dicalcium silicate (belite)	Ca_2SiO_4	$2\text{CaO} \cdot \text{SiO}_2$	C2S
Tricalcium aluminate	$\text{Ca}_3\text{Al}_2\text{O}_4$	$3\text{CaO} \cdot \text{Al}_2\text{O}_3$	C3A
Tetracalcium aluminoferrite	$\text{Ca}_4\text{Al}_n\text{Fe}_{2-n}\text{O}_7$	$4\text{CaO} \cdot \text{Al}_n\text{Fe}_{2-n}\text{O}_3$	C4AF

CaCO_3 (limestone)
 $2\text{SiO}_2 \cdot \text{Al}_2\text{O}_3$ (clay, shale)
 Fe_2O_3 (iron oxide)
 SiO_2 (silica sand)



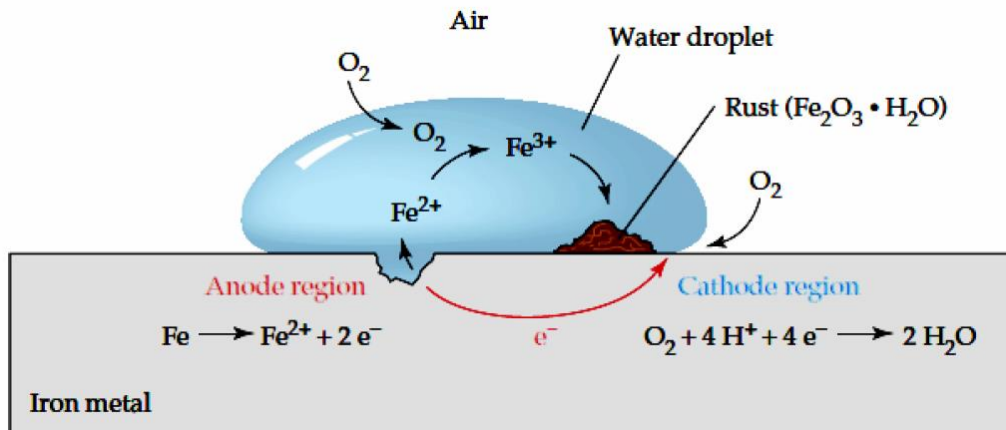


Rusting of Iron



Iron corrosion costs about 1% of the world's economy. Consists of the formation of hydrated oxide, $\text{Fe}(\text{OH})_3$ or $\text{Fe}(\text{OH})$. It is an electrochemical process which requires presence of (a) H_2O , (b) O_2 and (c) an electrolyte

If one of the conditions is not fulfilled, rusting does not occur to any significant extent. In air a relative humidity of 50% provides the necessary amount of water. It can be summarized as follows:



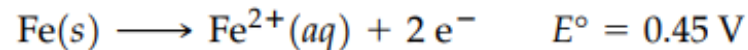
An electrochemical mechanism for corrosion of iron. The metal and a surface water droplet constitute a tiny galvanic cell in which iron is oxidized to Fe^{2+} in a region of the surface (anode region) remote from atmospheric O_2 and O_2 is reduced near the edge of the droplet at another region of the surface (cathode region). Electrons flow from anode to cathode through the metal, while ions flow through the water droplet. Dissolved O_2 oxidizes Fe^{2+} further to Fe^{3+} before it is deposited as rust



Iron is oxidized more readily in some regions (anode regions) than in others (cathode regions) because the composition of the metal is somewhat **inhomogeneous** and the **surface is irregular**. Factors such as impurities, phase boundaries, and mechanical stress may influence the ease of oxidation in a particular region of the surface.



At an anode region, iron is oxidized to Fe^{2+} ions,

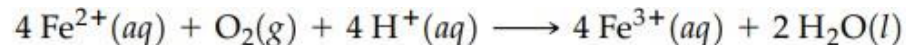


while at a cathode region, oxygen is reduced to water:

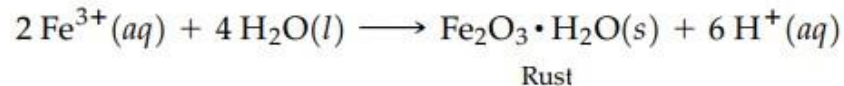


The electrons required for reduction of O_2 at the cathode region are supplied by a current that flows through the metal from the more easily oxidized anode region. The electrical circuit is completed by migration of ions in the water droplet.

When Fe^{2+} ions migrate away from the pitted anode region, they come in contact with O_2 dissolved in the surface portion of the water droplet and are further oxidized to Fe^{3+} ions:



Iron(III) forms a very insoluble hydrated oxide even in moderately acidic solutions, and the iron(III) is deposited as the familiar red-brown material that we call rust:



An electrochemical mechanism for corrosion also explains nicely why automobiles rust more rapidly in parts of the country where road salt is used to melt snow and ice. Dissolved salts in the water droplet greatly increase the conductivity of the electrolyte, thus accelerating the pace of corrosion.

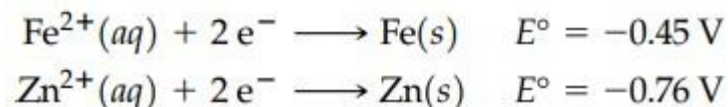
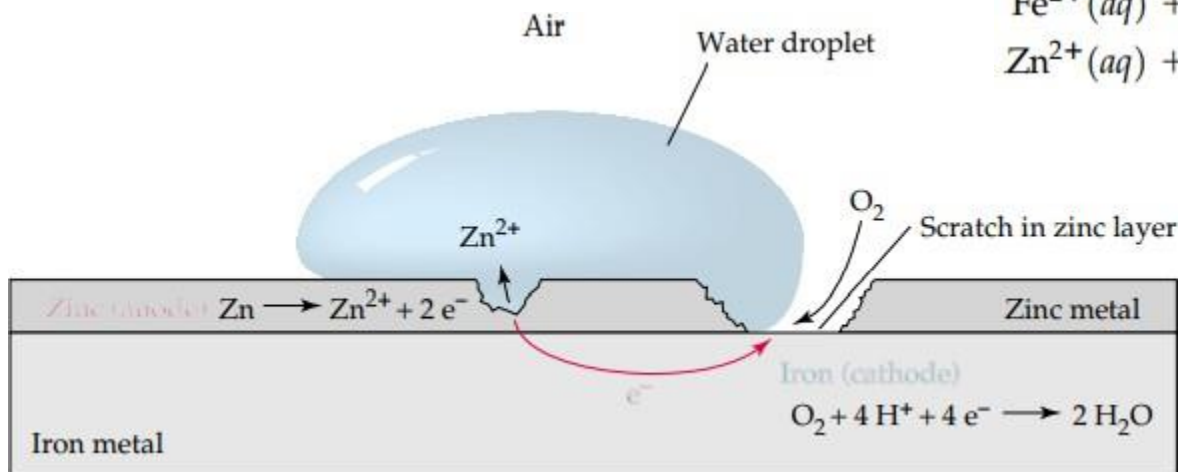


Prevention of Corrosion



A coat of paint is effective for a while, but rust begins to form as soon as the paint is scratched or chipped. Metals such as chromium, tin, or zinc afford a more durable surface coating for iron. The steel used in making automobiles, for example, is coated by dipping into a bath of molten zinc, a process known as **galvanizing**.

For example zinc is oxidized more easily than iron, and therefore, when the metal is oxidized, zinc is oxidized instead of iron. Any incipient oxidation of iron would be reversed immediately because Zn can reduce Fe^{3+} to Fe. As long as the zinc and iron are in contact, the zinc protects the iron from oxidation even if the zinc layer becomes scratched.



The zinc (anode), iron (cathode), and water droplet (electrolyte) constitute a tiny galvanic cell. Oxygen is reduced at the cathode, and zinc is oxidized at the anode, thus protecting the iron from oxidation.

The technique of protecting a metal from corrosion by connecting it to a second metal that is more easily oxidized is called **cathodic protection**.

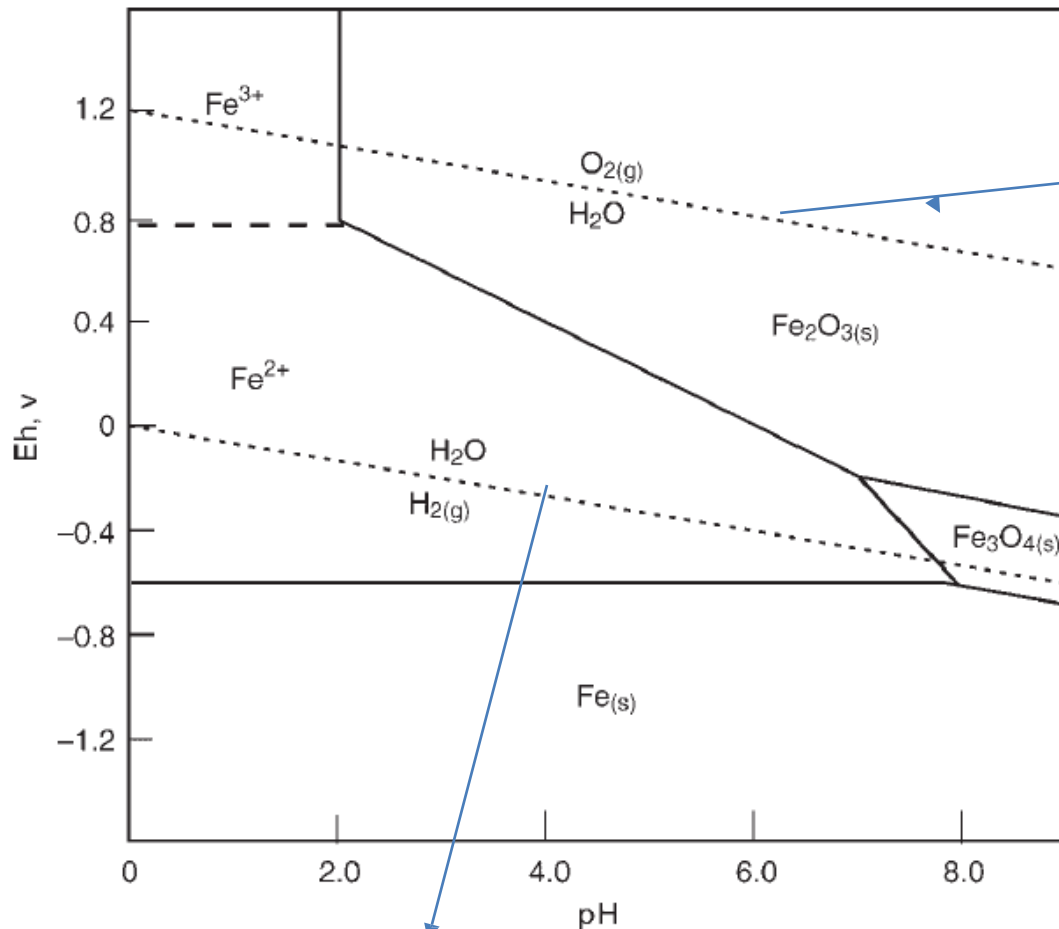


Eh–pH Diagram (Pourbaix diagram) for the Fe–H₂O System

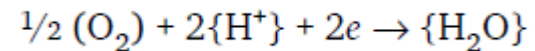


- Consider an iron bar immersed in water. Its fate under various oxidation/reduction potentials and pH values of the solution.
- When the pH of the solution is low and the oxidation potential of the iron bar is raised, iron will dissolve into the solution in the forms of either Fe²⁺ or Fe³⁺.
- As the pH of the solution is also increased, the surface iron will be subjected to oxidation to form either Fe₃O₄ or Fe₂O₃.
- It should be noted that the solid forms of Fe(OH)₂ and Fe(OH)₃ are thermodynamically less stable than Fe₃O₄ or Fe₂O₃.
- Therefore, only the following species will be considered in this study:
 - <Fe>, {Fe²⁺>, {Fe³⁺>, <Fe₃O₄>, and <Fe₂O₃>. The values of the Gibbs standard free energy of formation for these species are, respectively, 0, –20.3, –2.5, –242.4, and –177.1 kcal/mol; the value for water is 56.69 kcal/mol.

Eh-pH diagram for the system of iron-water at 25°C



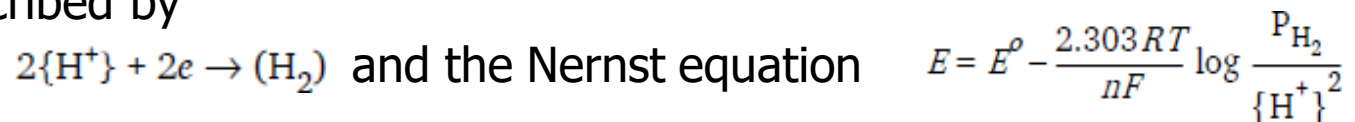
The upper boundary of the water stability line is presented by:



The Nernst equation for this relationship becomes

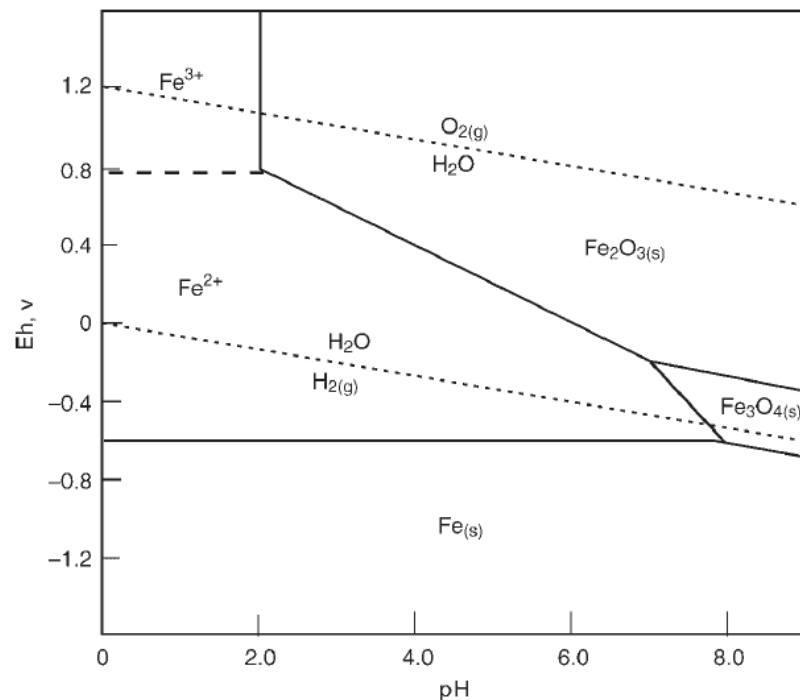
$$E = 1.23 - 0.059 \text{ pH} + 0.0148 \log p_{\text{O}_2}$$

The lower phase boundary of water in relation to the oxidation/reduction potential can be described by

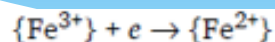


Which becomes

$$E = E^\circ - 0.059 \text{ pH} - 0.0295 \log p_{\text{H}_2}$$



The various equations represent the phase boundary lines for the Fe-bearing species



$$E = 0.771 - 0.059 \log \frac{\{\text{Fe}^{2+}\}}{\{\text{Fe}^{3+}\}}$$

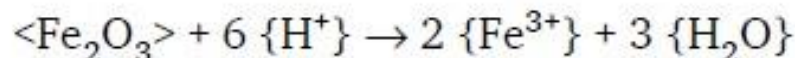
The activity of Fe^{2+} is the same as that of Fe^{3+} , $E = 0.771$ v. i.e., when the oxidation potential is > 0.771 v against the SHE, Fe^{3+} will be predominating, whereas Fe^{2+} will be more abundant when $E < 0.771$ v.

$$E = -0.440 + 0.0295 \log \{\text{Fe}^{2+}\}$$

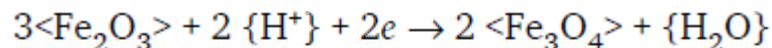
On the other hand, for $\{\text{Fe}^{3+}\} + 3e \rightarrow \langle \text{Fe} \rangle$,

$$E = -0.0366 + 0.0197 \log \{\text{Fe}^{3+}\}$$

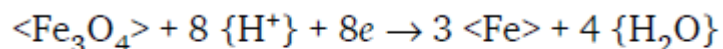
Now the phase boundaries for $\{\text{Fe}^{3+}\}/\langle \text{Fe}_2\text{O}_3 \rangle$ and $\{\text{Fe}^{2+}\}/\langle \text{Fe}_2\text{O}_3 \rangle$ should be identified:



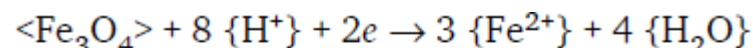
$$E = 0.728 - 0.1773 \text{ pH} - 0.059 \log \{\text{Fe}^{2+}\}$$



$$E = 0.221 - 0.059 \text{ pH}$$



$$E = -0.085 - 0.059 \text{ pH}$$



$$E = 0.98 - 0.2364 \text{ pH} - 0.0886 \log \{\text{Fe}^{2+}\}$$